

Horizon A6 Implemented Policies

Gate-owned maritime policy authorization over A5 physical assurance

Technical note based on the current Horizon implementation | September 2026

Purpose. A6 is a restrictive authorization layer over A5. A5 first selects a physically safe and recoverable command. A6 then determines whether that exact command is supported by the configured operational policy and trusted evidence. A6 may preserve or reduce A5 authority, but it cannot make an unsafe A5 decision safe.

$$\text{NormalDispatch}(a) = S_{A5}(a, x, \epsilon) \wedge P_{A6}(a, e, \pi) \wedge G(a), \quad (1)$$

where S_{A5} is physical-safety approval, P_{A6} is policy authorization using evidence e and bundle π , and G is the exclusive gate's identity, freshness, expiry and integrity validation.

1. Bounded operational scope

A6 currently evaluates a clear-visibility, daylight engineering-policy slice. Normal authorization requires the declared profile below.

Field	Required value	Field	Required value
Jurisdiction	Singapore	Service	Government, non-commercial
Vessel type	Power-driven	Length	Greater than 0 and under 50 m
Gross tonnage	At least 0 and under 300 GT	Passengers	None
Towing	False	Hazardous cargo	False
Supervision	Remote supervisor available	Conditions	Daylight, clear visibility
Narrow channel	Explicitly false	Traffic separation scheme	Explicitly false

The current development profile is a 12 m, 40 GT vessel. Restricted visibility, unknown context, unsupported contact classes, narrow channels, and traffic-separation schemes withhold normal authority. Supported encounters are power-driven vessels in sight.

2. Implemented navigation-rule proxies

Rule 5 – Lookout

A6 requires fresh evidence that all declared lookout capabilities are available:

$$V_{\text{visual}} \wedge V_{\text{auditory}} \wedge V_{\text{radar}} \wedge V_{\text{supervisor}} \wedge V_{\text{fresh}}.$$

If any value is false, missing, or stale, the lookout proxy is not satisfied or unknown and normal authority is withheld. This checks availability, not lookout effectiveness.

Rule 6 – Safe speed

The exact A5-selected speed must remain within the configured limit, traffic assessment must be available, and the vessel must be able to stop within clear distance:

$$v_{\text{cmd}} \leq v_{\text{max}}, \quad d_{\text{stop}} + d_{\text{reserve}} \leq d_{\text{clear}}. \quad (2)$$

For command-independent live evidence, A6 binds stopping distance conservatively:

$$d_{\text{stop}} = v t_{\text{response}} + \frac{v^2}{2a_{\text{min}}}, \quad v = \max(v_{\text{current}}, v_{\text{cmd}}). \quad (3)$$

The development bundle uses $v_{\text{max}} = 4 \text{ m/s}$ and $d_{\text{reserve}} = 20 \text{ m}$. These are configurable engineering thresholds, not universal legal definitions of safe speed.

Rule 7 – Risk of collision

For every supported contact, A6 checks that collision-risk evidence exists and that doubt is treated conservatively:

$$\text{risk doubt} \implies \text{treated as collision risk.}$$

If doubt is declared but the encounter is not treated as risk, Rule 7 is not satisfied. Missing risk evidence produces an unknown result.

Rule 8 – Action to avoid collision

Where collision risk is identified, action must be sufficiently early, substantial, and detectable. The development thresholds require

$$t_{\text{lead}} \geq 30 \text{ s} \quad (4)$$

and either

$$|\Delta\psi| \geq 20^\circ \quad \text{or} \quad \Delta v_{\text{reduction}} \geq 1 \text{ m/s.} \quad (5)$$

A6 derives course change, speed reduction and detectability from the actual command selected by A5. It overrides any external action claim.

Rule 13 – Overtaking

The encounter is classified as overtaking when ownship approaches from more than the configured abaft-the-beam angle:

$$|\beta_{\text{contact view}}| > 112.5^\circ. \quad (6)$$

Ownship is assigned the give-way duty, and its action must pass the early, substantial and detectable tests.

Rule 14 – Head-on situation

A head-on proxy requires relative bearing within 10° of directly ahead and course difference within 15° of reciprocal courses. A 2° ambiguity band protects classification boundaries. Ownship is assigned give-way duty and must make a positive course change, representing the expected starboard action, while also satisfying the Rule 8 thresholds.

Rule 15 – Crossing situation

A contact on the starboard side assigns ownship the give-way duty. A contact on the port side assigns ownship the stand-on duty. Classification uses relative bearing and withholds authorization when geometry is incomplete or within the ambiguity band.

Rule 16 – Action by give-way vessel

When ownship is the give-way vessel, the current proxy requires

$$(t_{\text{lead}} \geq 30) \wedge (|\Delta\psi| \geq 20^\circ \vee \Delta v_{\text{reduction}} \geq 1) \wedge \text{detectable.} \quad (7)$$

The quantities use seconds, degrees and metres per second respectively.

Rule 17 – Action by stand-on vessel

For the supported stand-on slice, A6 checks that course and speed are maintained. The command is treated as maintaining them when approximately

$$|\Delta\psi| < 1^\circ, \quad |\Delta v| < 0.1 \text{ m/s.} \quad (8)$$

This proxy does not model every condition in which a stand-on vessel may or must intervene.

3. Encounter classification

Encounter	Classification evidence	Assigned ownship duty	Rules
Overtaking	Contact view beyond 112.5° abaft beam	Give way	R13, R16
Head-on	Near-zero bearing and near-reciprocal courses	Give way	R14, R16
Crossing, contact starboard	Positive relative bearing	Give way	R15, R16
Crossing, contact port	Negative relative bearing	Stand on	R15, R17

Incomplete or boundary-ambiguous geometry is classified as unknown. A6 withholds normal authority rather than guessing the encounter duty.

4. Gate and evidence policies

- **A5 safety dominates.** Only valid A5 `pass` or `modify` decisions with an issued command are eligible for normal authorization.
- **Exact-command binding.** A6 evaluates the command A5 selected, not the original decision-AI proposal.
- **Trusted evidence only.** Evidence comes from a gate-configured adapter, not from the decision AI.
- **Recorded provenance.** Operational bundles require recorded provenance, a runtime-authority source pin, and source-artifact hash validation.
- **Identity binding.** Run, branch, tick, snapshot hash, A5 decision and command hash must agree.
- **Freshness and expiry.** Authorization expires at the earliest A5 decision, cycle deadline, evidence, or policy-bundle expiry.
- **Fail closed.** Missing, malformed, mismatched, stale, unsupported, unresolved or late evidence causes withholding.
- **Compute budget.** The default evaluation budget is 10 ms; exceeding it causes withholding.
- **No authority expansion.** A6 cannot authorize a physically impermissible command.

5. Withholding and recovery

When A6 withholds a normal command, the gate attempts a fresh stored recovery and rechecks it against the current A5 input. If no valid stored recovery exists, the gate derives its canonical minimum-risk command and marks assurance unknown.

A5 recovery, minimum-risk, recovery-latch and watchdog actions may retain emergency safety authority under `A6_EMERGENCY_POLICY_OVERRIDE`. This is explicitly not A6 policy authorization.

$$a_{\text{dispatch}} = \begin{cases} a_{\text{A5}}, & \text{A5 safe and A6 authorized,} \\ a_{\text{recovery}}, & \text{normal authority withheld and recovery valid,} \\ a_{\text{minimum risk}}, & \text{otherwise, with assurance unknown.} \end{cases} \quad (9)$$

6. Current exclusions

Not a complete legal-compliance system. A6 implements bounded engineering proxies for Rules 5–8 and 13–17. It does not establish full COLREG, Singapore regulatory, vessel-certification, or operational compliance.

The current implementation does not cover:

- Rule 2 general responsibility;
- Rule 9 narrow channels or Rule 10 traffic-separation schemes;
- Rule 12 sailing vessels, Rule 18 vessel responsibilities, or Rule 19 restricted visibility;
- complete multi-contact rule conflict resolution;
- auditory sensor interpretation or complete live vessel-policy feeds;
- weapons, targeting, classified doctrine, rules of engagement, or naval mission policy;
- alternative-manoeuve search after a policy veto.

7. Development parameters

Policy parameter	Development value
Maximum clear-visibility speed	4 m/s
Stopping-distance reserve	20 m
Minimum early-action lead	30 s
Substantial course change	20°
Substantial speed reduction	1 m/s
Head-on bearing tolerance	10°
Reciprocal-course tolerance	15°
Classification ambiguity band	2°
Overtaking abaft-beam boundary	112.5°

These are versioned development settings. Operational values require a recorded, hash-pinned policy source, applicability review and deployment-specific approval.

Implementation references

- `services/assurance/horizon_assurance/policy_shadow.py` — scope, classification, findings and bundle validation;
- `services/assurance/horizon_assurance/policy_enforcement.py` — evidence binding, authorization, expiry and fail-closed enforcement;
- `services/assurance/docs/a6-policy-enforcement.md` — configuration, command behavior and limitations;
- `experiment/manifests/a6-a32-enforcement-development.json` — development thresholds and vessel profile.

Interpretation note. A *satisfied* finding means that the configured engineering predicate passed using the supplied bounded evidence. It is not a legal judgment and must not be reported as unrestricted COLREG compliance.